Switched-capacitor power converters
Lecture I - Homework

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Homework – Exercise 1:

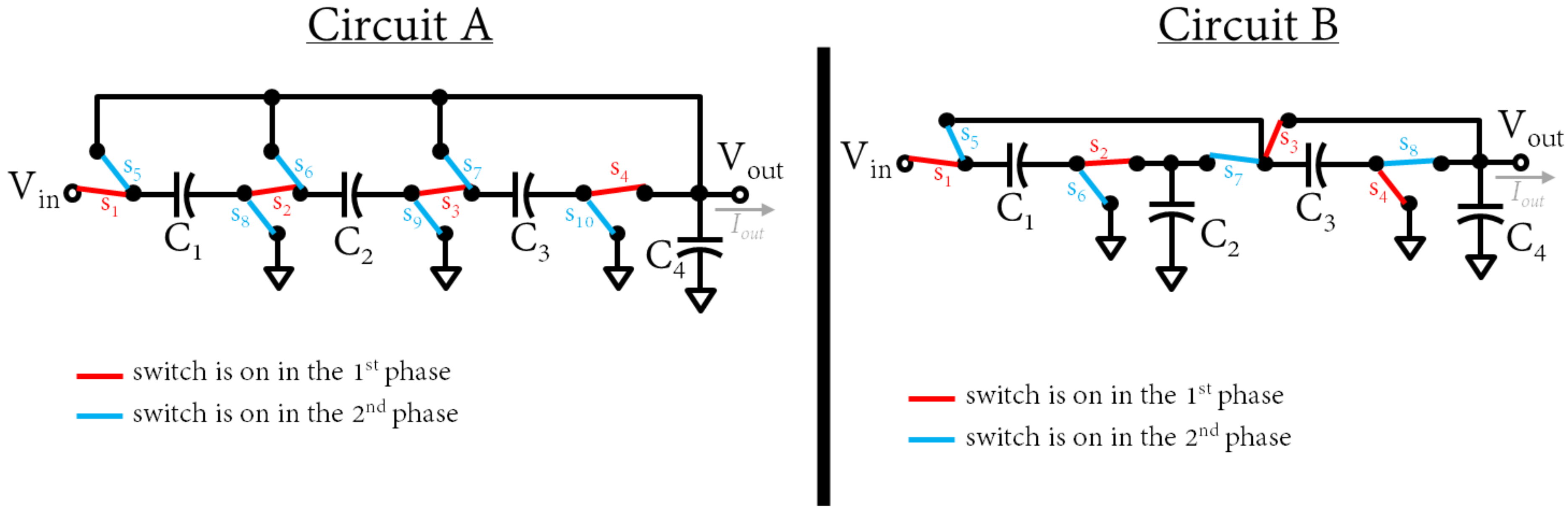

Figure 1

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Homework – Exercise 1:

1. Analyse the switched-capacitor power converters from Figure 1A. Assume that C_4 is large enough so that the small-ripple assumption holds:

   a) What is the conversion ratio of this power converter?

   b) What is its R_{FSL} output impendance as a function of the switches resistance (R_1,R_2,R_3,... R_{10})?

   c) Create the corresponding a_R vector. What would be its p value if all the switches would have the same R_{on}?

   d) Find the R_{SSL} output impendance as a function of the floating capacitors values (C_1,C_2 and C_3). If C_1=C_2=C_3, what is the m value of these topologies?

   e) Given a certain amount of total floating capacitance (C_T), what is the distribution of capacitance among C_1, C_2 and C_3 that would result in the lowest R_{out}? What is the m value in such case?

   f) Build the circuit with ideal capacitors and switches, consider the same R_{on}=5Ω for all the switches, a total amount of floating capacitance C_T=4.5nF, and fs=5MHz. Reproduce the R_{out}=$f(f_s)$ curve considering an optimum distribution of the floating capacitance. Consider the points below:

      - Use the following values V_{in}=1V and I_{out}=1mA.
      - Make sure that the non-overlapping time plus the fall/rise times of the switching signals is ≤2% of the switching period.

2. Find the conversion ratio M, the R_{FSL} and the R_{SSL} for the circuit in Figure 1B. What is the m value in case of an optimum distribution of the available capacitance C_T? Compare the results with the obtained for Figura1A. Which circuit provides the lowest output impedance? What is the main reason behind this?

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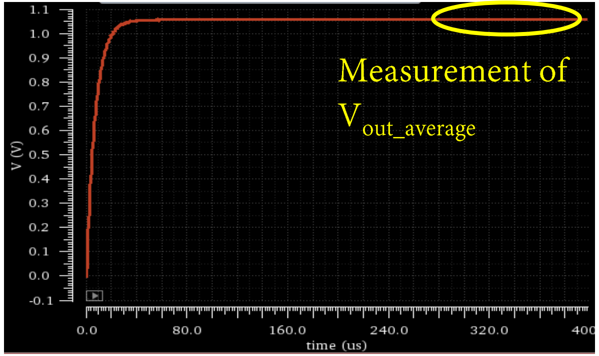
Homework – Exercise 1:

Tips for the simulations:

- In the simulation of the SCPC, use ideal switches (with the calculated R_{on}) and ideal capacitors.
- In order to help the convergence of the simulator, add small parasitic capacitances between the terminals of the capacitors and Gnd. In general, *1fF* should be enough (a too big value will distort the operation of the converter).
- Measure V_{out} when the signal has completely settled (allow ~10 time constants of the equivalent *RC* circuit of the output node).

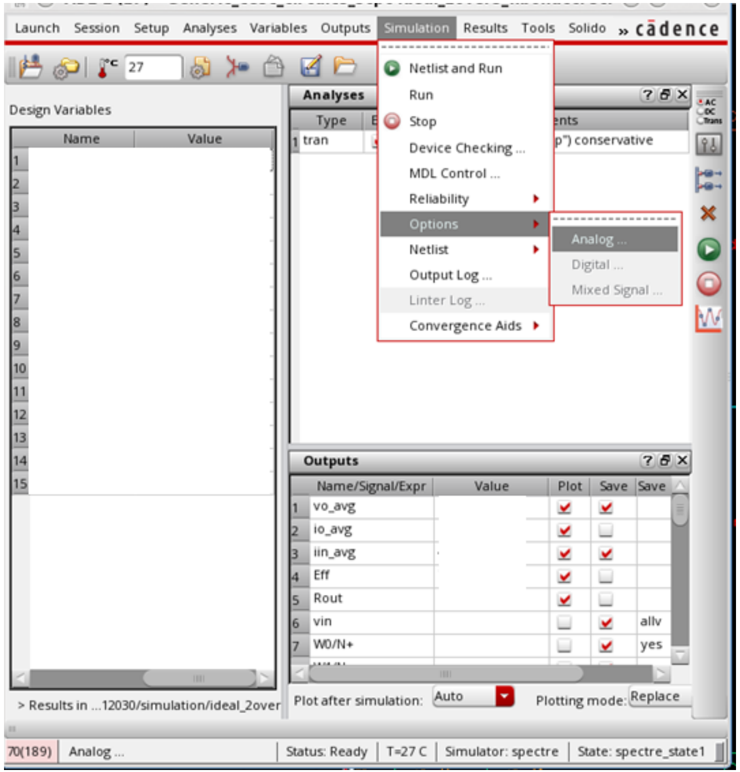


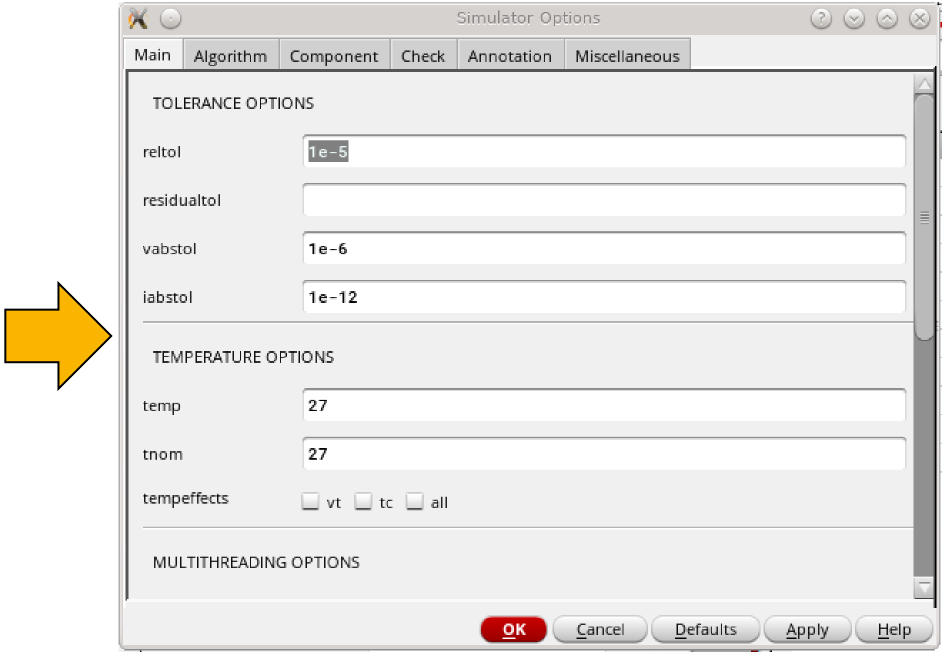
- To measure the average value of V_{out}, clip the signal for a large number of switching cylces (~64). This will increase the accuracy of the measurement.
- Implement the non-overlapping signals with the provided voltage sources.

Homework – Exercise 1:

<u>Tips for the simulations:</u>

- Increase the accuracy settings of the simulator:
  - Conservative → Tick the corresponding box when defining the transient simulation.
  - reltol=1e-5, vabstol=1e-6 → These parameters can be found in *Simulation* → *Options* → *Analog…* (as shown below)





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Homework – Exercise 1:

This implementation for Vctrl1 and Vctrl2 allows an easy implementation of non-overlapping signals

Power switch

Property	Value	Display
Library Name	analogLib	off
Cell Name	switch	off
View Name	symbol	off
Instance Name	W7	off

Add Delete Modify

CDF Parameter	Value	Display
Open voltage	0 V	off
Closed voltage	1 V	off
Open switch resistance	100M Ohms	off
Close switch resistance	R Ohms	off
Multiplier		off
Estimated operating region		off

Floating Capacitors:

- They are ideal capacitors.
- Provide them an initial condition to speed up the transient a bit (but it doesn't really matter)
- Connect a couple of additional capacitors between their terminals and ground:
  - To make the convergence of the simulation easier.
  - Afterwards will easily allow the inclusion of the parasitics of the floating caps (not needed for Homework of Lecture 1)

Vctrl1 → phase 1 clock

Property	Value	Display
Library Name	analogLib	off
Cell Name	vpulse	off
View Name	symbol	off
Instance Name	V1	off

Add Delete Modify

User Property	Master Value	Local Value	Display
lvsIgnore	FALSE		off

CDF Parameter	Value	Display
Change Tool Filter...		
Frequency name for 1/period		off
Noise file name		off
Number of noise/freq pairs	0	off
DC voltage		off
AC magnitude		off
AC phase		off
XF magnitude		off
PAC magnitude		off
PAC phase		off
Voltage 1	0 V	off
Voltage 2	1 V	off
Period	1/f s	off
Delay time	0 s	off
Rise time	tfr s	off
Fall time	tfr s	off
Pulse width	0.5/f-tnonov-2*tfr s	off

Vctrl2 → phase 2 clock

Property	Value	Display
Library Name	analogLib	off
Cell Name	vpulse	off
View Name	symbol	off
Instance Name	V2	off

Add Delete Modify

User Property	Master Value	Local Value	Display
lvsIgnore	FALSE		off

CDF Parameter	Value	Display
Change Tool Filter...		
Frequency name for 1/period		off
Noise file name		off
Number of noise/freq pairs	0	off
DC voltage		off
AC magnitude		off
AC phase		off
XF magnitude		off
PAC magnitude		off
PAC phase		off
Voltage 1	0 V	off
Voltage 2	1 V	off
Period	1/f s	off
Delay time	0.5/f s	off
Rise time	tfr s	off
Fall time	tfr s	off
Pulse width	0.5/f-tnonov-2*tfr s	off

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